Study of Geiger–Müller (GM) Counter

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The characteristics and performance of a Geiger–Müller (GM) counter were studied using β and γ sources. The plateau region was identified in the range 360–600 V with percentage slopes of $(7.8 \pm 1.4)\%$ for γ and $(11.6 \pm 2.3)\%$ for β, and an operating voltage of 480 V was calculated. The inverse square law was verified with a fitted exponent $n = -1.49 \pm 0.07$, showing reasonable agreement with theory. The detector efficiencies were determined as $(5.47 \pm 0.69)\%$ for β and $(1.34 \pm 0.09)\%$ for γ radiation. Statistical analysis of background counts yielded $\bar{N} = 70.56$, $\sigma = 9.41$, confirming Poisson nature ($\mu \approx \sigma^2$), while higher count rates followed a Gaussian distribution with normalized parameters $\mu = 0.32 \pm 0.04$ and $\sigma = 1.00 \pm 0.04$. Using the half-thickness method, the endpoint energy of Sr-90 was estimated as 1.80 ± 0.52 MeV with a corresponding range of 0.84 ± 0.27 g/cm^2. Back-scattering of β particles showed an increasing trend with absorber thickness. Bremsstrahlung studies confirmed that radiation intensity increases with atomic number. Overall, the experimental results are in good agreement with theoretical expectations, demonstrating radiation detection, statistical nature of radioactive decay, and interaction of radiation with matter.

I. OBJECTIVE

1. To study the variations of count rate with applied voltages and thereby determine the plateau the operating voltage and the slope of the plateau.
2. To verify the inverse square law using a Gamma source.
3. Estimate the efficiency of the G.M. detector for a Gamma source and a beta source.
4. Study the nuclear counting statistics.
5. To study determination of beta particle range and maximum energy using the half thickness method.
6. To study the back-scattering of beta particles.
7. To study the production and attenuation of Bremsstrahlung.

II. THEORY

A. Construction of Geiger–Müller Counter

A Geiger–Müller (GM) counter is a gas-filled radiation detector used to detect ionizing radiation. It consists of a cylindrical metal tube acting as the cathode and a thin wire stretched along its axis serving as the anode (see Fig. 1). The tube is filled with an inert gas such as argon at low pressure, along with a small quantity of quenching gas (such as alcohol vapour or halogen gas).

A high voltage (typically 400–900 V) is applied between the anode and cathode. The central wire is maintained at a positive potential relative to the cylindrical wall [1].

B. Working Principle

The operation of the GM counter is based on the ionization of gas molecules by incident radiation. When radiation enters the tube, it produces primary electron-ion pairs. Due to the strong electric field near the anode wire, the electrons accelerate and cause further ionization, resulting in a Townsend (see Figure 2). This leads to a large number of charge carriers and produces a detectable current pulse.

The pulse generated is independent of the initial energy of the incident radiation, which makes the GM counter suitable for counting purposes but not for energy measurement.

A small amount of quenching gas is added to terminate the discharge. It absorbs the energy of positive ions and prevents continuous ionization, allowing the detector to return to its normal state after each event.

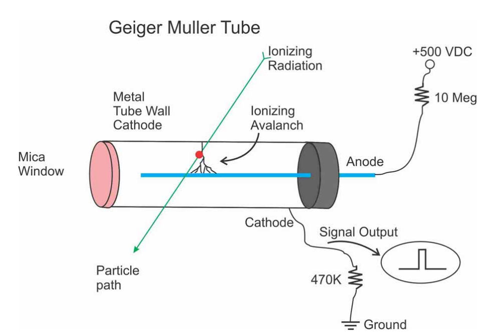

FIG. 1. A schematic diagram of Geiger Muller counter

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After each discharge, there is a short interval during which the detector is insensitive to incoming radiation. This occurs because positive ions formed during the avalanche take time to drift away from the anode, temporarily reducing the electric field strength. As a result, if radiation events occur during this , called the *dead time*, they are not recorded, leading to an underestimation of the true count rate at high intensities.

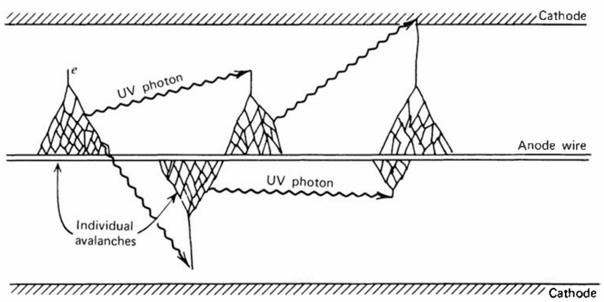

FIG. 2. The mechanism by which additional avalanches are triggered in a Geiger discharge.

C. Characteristics of GM Counter

The characteristic curve of a GM counter shows the variation of count rate with the applied high voltage (EHT). At sufficiently high voltage, the counter operates in the *Geiger region*, where each ionizing event produces a complete avalanche, resulting in pulses of identical magnitude. In this region, the count rate becomes nearly independent of the applied voltage over a range, known as the *Geiger plateau*. The operating voltage of the GM counter is chosen within this plateau to ensure stable (see Figure 3).

If the applied voltage is increased beyond this region, the detector enters the *continuous discharge region*, where ionization occurs continuously even without incident radiation. In this condition, the counter gives excessively high counts and becomes unsuitable for measurements.

D. Inverse Square Law

The intensity of radiation from a point source decreases with distance according to the inverse square law:

$$I \propto \frac{1}{r^2}$$

This is because radiation spreads uniformly over the surface of a sphere of radius r, whose area increases as $4\pi r^2$.

In the experiment, the count rate measured by the GM counter is proportional to the intensity of radiation. By varying the distance between the source and the detector and correcting for background radiation, the inverse square dependence can be verified.

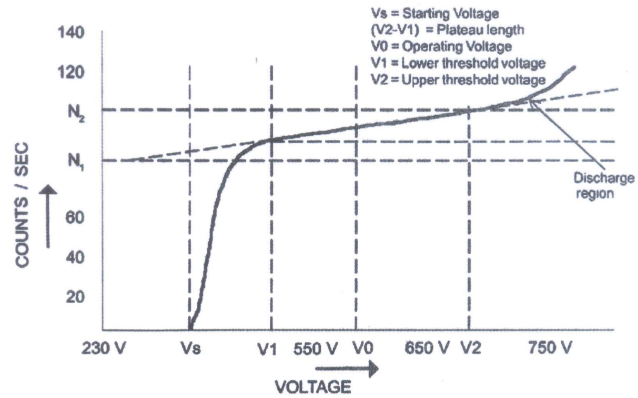

FIG. 3. Typical GM Characteristics

E. Efficiency of GM Detector

The efficiency of a Geiger–Müller (GM) detector is defined as the ratio of the number of counts recorded by the detector to the actual number of radiation quanta emitted by the source. It is expressed as:

$$\text{Efficiency} = \frac{\text{CPS}}{\text{DPS}} = \frac{N}{R}$$

where CPS = Counts per second and DPS = Disintegrations per second falling on the window of the detector.

The efficiency depends on several factors such as the geometry of the setup, distance between the source and detector, absorption in air and window material, and the intrinsic detection capability of the GM tube.

Since not all emitted particles reach or are detected by the GM counter, the efficiency is always less than unity. Determination of detector efficiency is important for accurate measurement of radioactive source strength.

F. Nuclear Counting Statistics

Radioactive decay is a random process, and the number of disintegrations occurring in a given time interval follow a Poisson distribution. If λ is the average count in a fixed time interval, the probability of observing n counts is given by the Poisson distribution:

$$P(n) = \frac{\lambda^n e^{-\lambda}}{n!} \tag{1}$$

The mean and variance of the distribution are: $\langle n \rangle = \lambda, \quad \sigma^2 = \lambda$, and hence the standard deviation is: $\sigma = \sqrt{\lambda}$

For large values of λ, the Poisson distribution approaches a Gaussian (normal) distribution:

$$P(n) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(n-\lambda)^2}{2\sigma^2}\right] \tag{2}$$

The statistical nature of radioactive decay is verified by comparing the experimental distribution of counts with the theoretical Poisson (or Gaussian) distribution.

G. Range of β-Particles (Half Thickness Method)

The range of β-particles in a material is determined by studying the attenuation of radiation intensity as it passes through an absorber. As the thickness of the absorber increases, the count rate decreases due to energy loss of β-particles[2].

The intensity variation approximately follows an exponential law:

$$I = I_0 e^{-\mu t}$$

where μ is the absorption coefficient and t is the thickness of the absorber. In the half thickness method, the thickness $t_{1/2}$ is defined as the thickness required to reduce the intensity to half of its initial value: $I = \frac{I_0}{2}$

The range of beta particles is intrinsically linked to their maximum, or end-point, energy. The empirical relationship defining the range of beta particles is given by the equation

$$R_0 = (0.52\ E_0 - 0.09)\ g/cm^2 \tag{3}$$

where E_0 represents the end point energy of the beta rays emitted from the radioactive source, measured in MeV.

The ratio of the absorber thickness required to reduce the beta ray counts from one source to half its initial value ($t_1^{1/2}$) compared to the half-thickness required for a second source ($t_2^{1/2}$) is equal to the ratio of their respective ranges,

$$\frac{t_1^{1/2}}{t_2^{1/2}} = \frac{R_1}{R_2} \tag{4}$$

By determining these half-thickness values through successive absorption measurements, the unknown range and maximum energy of a specific beta-emitting source can be calculated.

H. Back-scattering of β-Particles

Back-scattering refers to the phenomenon in which incident particles are deflected back from the surface of a material due to interactions with atomic nuclei and electrons. When a β source is placed near a scattering material, some of the emitted particles are reflected back towards the detector.

The extent of back-scattering depends on the atomic number (Z) of the material. A higher atomic number in materials produces more back-scattering due to stronger interaction with the incident electrons.

I. Production and Attenuation of Bremstrahlung

Bremsstrahlung is secondary electromagnetic radiation produced when charged particles, such as beta particles, decelerate upon interacting with atomic nuclei. The intensity of this generated radiation is directly proportional to the atomic number (Z) of the absorbing material; materials with a higher atomic number produce significantly more bremsstrahlung.

III. APPARATUS

End-window GM tube, GM counting unit, High voltage supply, Counter timer, Source stand, Connecting cables, Aluminum sheets, Lead absorbers for shielding, Perspex sheet, Aluminum and Copper sheet, Beta sources (^{204}Tl or ^{90}Sr), Gamma sources (^{137}Cs or ^{60}Co).

IV. DATA ANALYSIS

A. GM Characteristics and Operating Voltage

The variation of count rate with applied high voltage (EHT) for the GM counter using a gamma source and beta source are shown in Fig. 4 and 5 respectively.

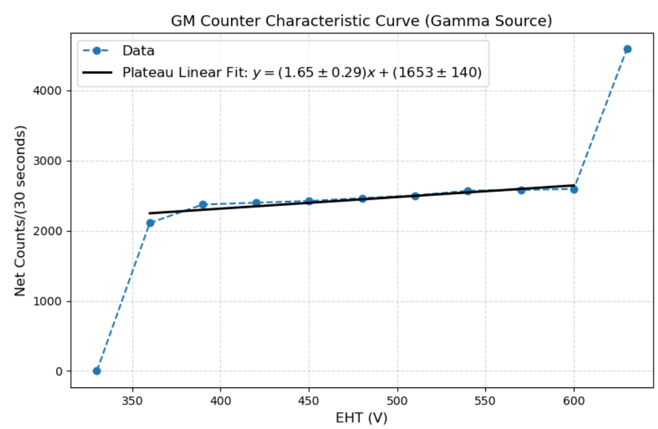

FIG. 4. GM counter characteristic curve for gamma source (Cs-137).

From both the plot, the plateau region is identified in the range 360 V to 600 V. A linear fit was performed in this region of the form:

$$y = mx + c$$

where the slope was obtained as

$$\text{For } \gamma \text{ source}, m_\gamma = 1.65 \pm 0.29$$
$$\text{For } \beta \text{ source}, m_\beta = 0.36 \pm 0.07$$

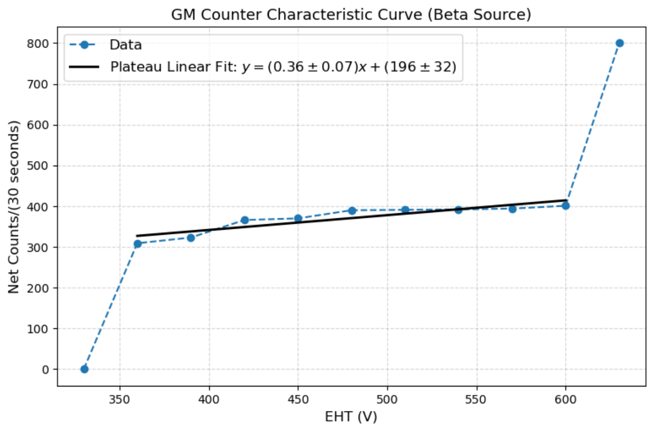

FIG. 5. GM counter characteristic curve for beta source (Tl-204).

The operating voltage is chosen at the middle of the plateau region (same for both sources):

$$V_{\text{op}} = (600 + 360)/2 = 480V$$

Also, the % slope of the plataeu was obtained using the given formula:

$$\begin{aligned}
\%\text{slope} &= \frac{N_2 - N_1}{N_1} \times \frac{100}{V_2 - V_1} \times 100\% \\
&= \frac{m \times 100}{N_1} \times 100\% \\
\%(\text{slope})_\gamma &= 7.8\% \\
\%(\text{slope})_\beta &= 11.6\%
\end{aligned}$$

B. Inverse Square Law

The variation of count rate R with distance d from the gamma source was studied to verify the inverse square law (see Figure 6). The experimental data was corrected for background and converted to count rate.

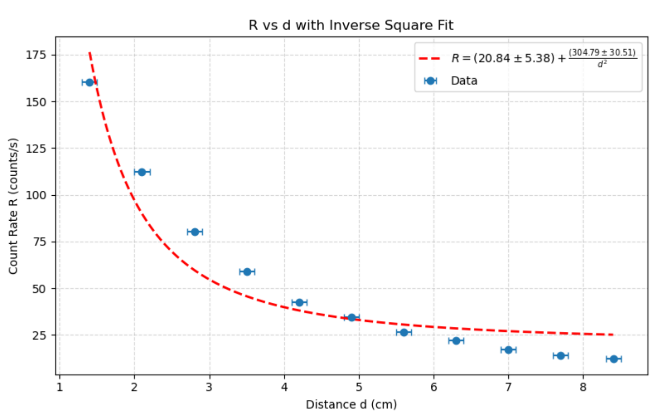

FIG. 6. R vs d with fit of the form $R = A + \frac{B}{d^2}$.

The data was fitted to the relation

$$R = A + \frac{B}{d^2}$$

using a non-linear form (using inearization in $1/d^2$). The fitted parameters were obtained as

$$A = 20.84 \pm 5.38 \tag{5}$$
$$B = 304.79 \pm 30.52 \tag{6}$$

The small value of A represents the background contribution, while B corresponds to source intensity.

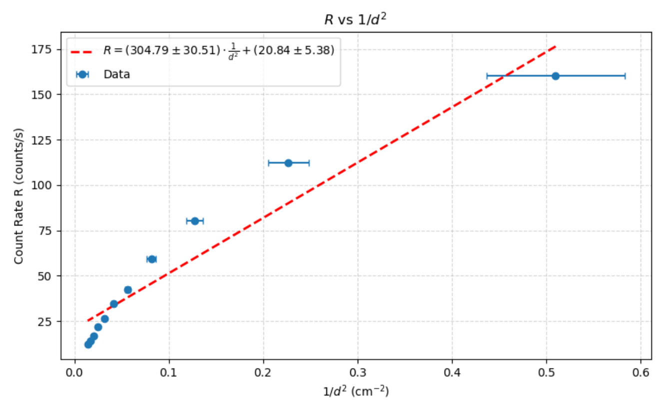

FIG. 7. R vs $1/d^2$ showing linear nature.

This nature can also be seen from the R vs $1/d^2$ plot linear fit as shown in Figure 7). The linearity of this plot confirms the inverse square dependence of the count rate on distance.

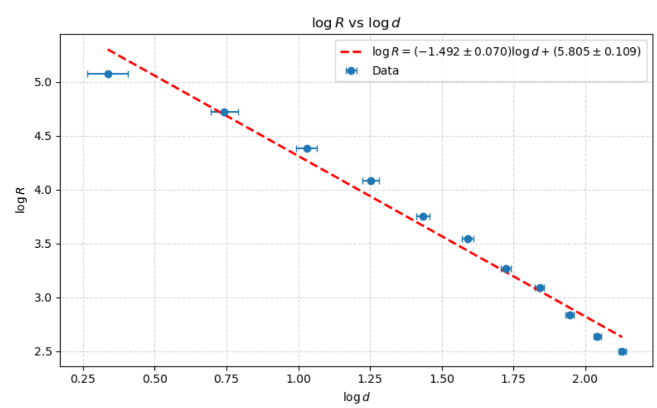

FIG. 8. $\log R$ vs $\log d$ plot.

A log-log plot was also analyzed using a linear fit of the (see Figure 8):

$$\log R = n \log d + k$$

The fitted slope and intercept were found to be

$$n = -1.49 \pm 0.07 \tag{7}$$
$$k = 5.808 \pm 0.109 \tag{8}$$

The slope is expected to be close to -2 for an ideal inverse square law. The error bars for X-axis marked in the Fig. 6, 7, 8 are corresponding to the uncertainty in distance from the source $\Delta d = 0.1$cm.

All three plots—non-linear, linearized, and log-log fitting—consistently indicate that the count rate varies almost inversely with the square of distance. The experimental value of the exponent is close to -2, thereby verifying the inverse square law for gamma radiation within experimental uncertainty.

C. Estimation of Efficiency of GM Detector

The efficiency of the GM detector for beta and gamma sources was estimated by comparing the measured count rate with the expected disintegration rate incident on the detector window.

The present activity of the sources was calculated using the radioactive decay law:

$$A = A_0 e^{-\lambda t} = A_0 e^{-\frac{\ln 2}{t_{1/2}} t}$$

where $t = 10$ years, since the activity of sources provided dated back to 2016.

$$A_{0\beta} = 10\text{kBq}, \qquad A_{0\gamma} = 79\text{kBq}$$

For the beta source Tl-204 ($t_{1/2} = 3.78$ years) and gamma source Cs-137 ($t_{1/2} = 30$ years), the present activities were obtained as:

$$A_\beta = 1.60\text{kBq}, \qquad A_\gamma = 62.7\text{kBq}$$

The number of disintegrations per second incident on the detector window is given by:

$$R = \frac{A \cdot \pi(d^2/4)}{4\pi D^2} = \frac{Ad^2}{16D^2}$$

where d is the diameter of the detector window and D is the distance of the source from the detector.

The efficiency of the detector is then calculated as:

$$\eta = \frac{N}{R} \times 100\%$$

where N is the net count rate recorded by the detector.

The disintegrations per second incident on detector window were calculated to be:

$$\text{For } \gamma \text{ source}, R_\gamma = 352.7$$
$$\text{For } \beta \text{ source}, \Delta R_\beta = 225.0$$

Using the above relations, the efficiencies were obtained as:

$$\eta_\beta = 5.47\% \tag{9}$$
$$\eta_\gamma = 1.34\% \tag{10}$$

The efficiency for beta particles is significantly higher than that for gamma rays due to their stronger interaction with the detector medium. The low efficiency for gamma rays arises from their weak interaction probability and high penetration power. The results are consistent with theoretical expectations and demonstrate the dependence of detector efficiency on the nature of the radiation and interaction mechanisms.

D. Study of Nuclear Counting Statistics

1. Background Counts

The statistical nature of background radiation was analyzed using multiple independent measurements. The mean count was calculated as:

$$\bar{N} = \frac{1}{n}\sum_{i=1}^{n} N_i$$

and the variance was obtained using:

$$\sigma^2 = \frac{1}{n-1}\sum_{i=1}^{n}(N_i - \bar{N})^2$$

The standard deviation is given by:

$$\sigma = \sqrt{\sigma^2}$$

The obtained values were:

$$\bar{N} = 70.56 \tag{11}$$
$$\sigma^2 = 88.52 \tag{12}$$
$$\sigma = 9.41 \tag{13}$$
$$\text{Points within } \pm\sigma = 41/68 \approx 60.3\% \tag{14}$$

It was observed that a good fraction of data points lie within $\bar{N} \pm \sigma$, consistent with statistical fluctuations (see Figure 9).

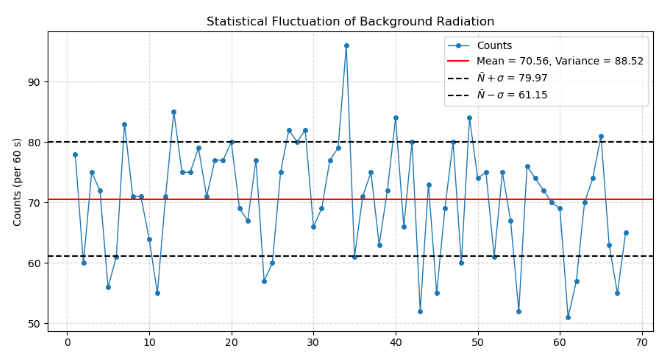

FIG. 9. Statistical fluctuation of background counts.

The frequency distribution of background counts was analyzed and fitted with a Gaussian function of the form (see Figure 10):

$$f(N) = A \exp\left(-\frac{(N-\mu)^2}{2\sigma^2}\right)$$

The fitted parameters of the Gaussian were obtained to be:

$$A = (11.4 \pm 1.3), \quad \mu = (72.75 \pm 1.27), \quad \sigma = (9.61 \pm 1.29).$$

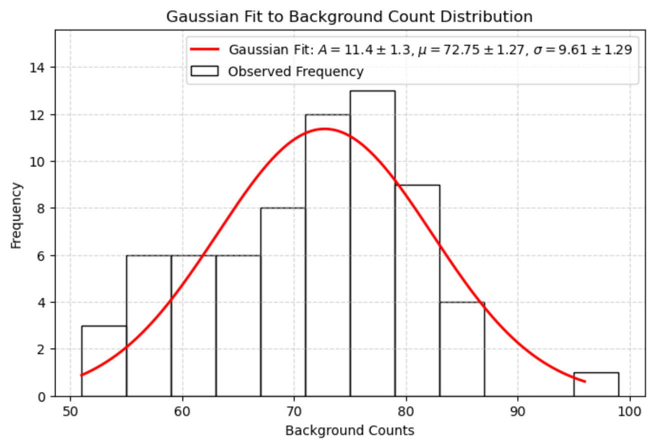

FIG. 10. Gaussian fit to background count distribution.

The agreement between μ and $\bar{N}$ and the bell-shaped distribution indicate that background radiation follows Poisson statistics, which approximates a Gaussian distribution for large number of counts.

2. *Counts with β Source*

For higher count rates (with β source), the statistical distribution was analyzed using deviations from the mean normalised by the standard deviation:

$$z = \frac{N_i - \bar{N}}{\sigma}$$

where σ is the standard deviation.

The values of z were rounded to the nearest half-integer and their frequency distribution was obtained shown in the Figure 11.

The mean and variance of the counts were:

$$\bar{N} = 734.55, \quad \sigma^2 = 700.86$$

Variance $\approx$ Mean, confirming Poisson nature of radioactive decay. The frequency distribution was fitted with a Gaussian function:

$$f(z) = A \exp\left(-\frac{(z-\mu)^2}{2\sigma^2}\right)$$

The fitted parameters were obtained to be:

$$\mu = (0.32 \pm 0.04), \quad \sigma = (1.00 \pm 0.04).$$

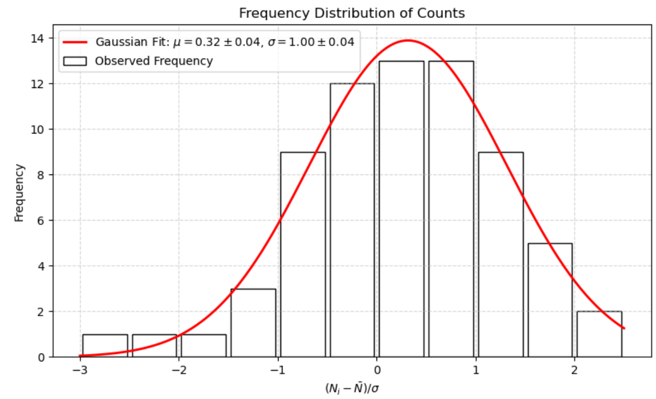

FIG. 11. Frequency distribution of normalized counts with Gaussian fit.

For low count rates (background), the distribution follows Poisson statistics with variance approximately equal to the mean. For higher count rates, the distribution becomes symmetric and closely follows a Gaussian form. The fitted parameters $\mu \approx 0$ and $\sigma \approx 1$ confirm the normalization and validate the expected statistical behavior of radioactive decay.

E. β particle Range and Endpoint energy using Half Thickness method

The attenuation of beta particles in aluminum was studied by measuring the count rate as a function of Aluminum absorber thickness for a known energy source (Tl-204) and a source with unknown endpoint energy (Sr-90).

1. *Half Thickness Method*

The half thickness $t_{1/2}$ is defined as the absorber thickness at which the count rate falls to half of its maximum value:

$$N(t_{1/2}) = \frac{N_0}{2}$$

The value of $t_{1/2}$ was determined graphically using linear interpolation between successive data points.

2. *Tl-204 Source*

From the graph, the half thickness was obtained as (see Figure 12):

$$t_{1/2}^{(\mathrm{Tl})} = 37.3\ \mathrm{mg/cm^2}$$

The range of beta particles is proportional to the half thickness:

$$R \propto t_{1/2}$$

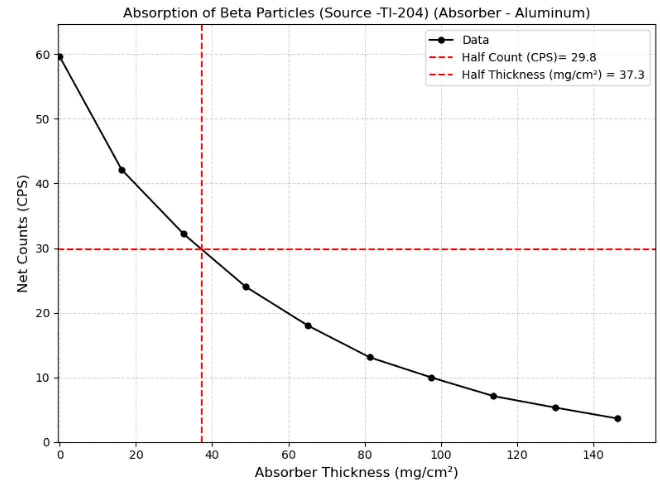

FIG. 12. Absorption curve for Tl-204 beta source.

Using the known endpoint energy of Tl-204, the corresponding range was used as a reference to calculate the range of unknown source.

3. Sr-90 Source

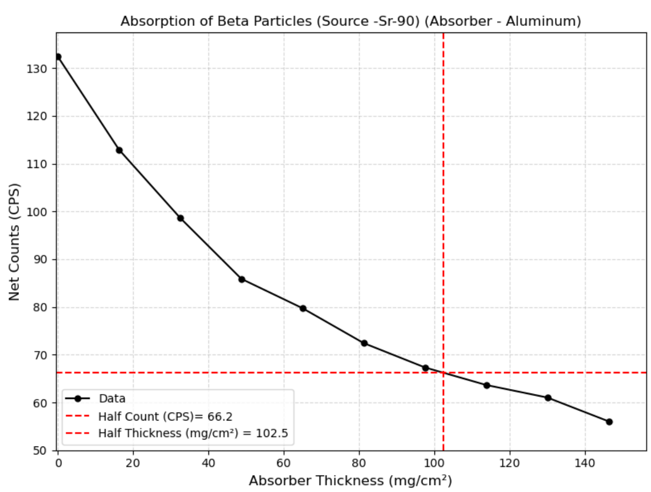

FIG. 13. Absorption curve for Sr-90 beta source.

The half thickness for Sr-90 was obtained as (see Figure 13):

$$t_{1/2}^{(\mathrm{Sr})} = 102.5 \text{ mg/cm}^2$$

Using the ratio of half thicknesses, the range of Sr-90 beta particles is calculated:

$$\frac{R_{\mathrm{Sr}}}{R_{\mathrm{Tl}}} = \frac{t_{1/2}^{(\mathrm{Sr})}}{t_{1/2}^{(\mathrm{Tl})}}$$

Thus,

$$R_{\mathrm{Sr}} = R_{\mathrm{Tl}} \cdot \frac{t_{1/2}^{(\mathrm{Sr})}}{t_{1/2}^{(\mathrm{Tl})}} = 0.8444 \mathrm{g}/cm^2$$

The endpoint energy (E_0) is related to the range by:

$$R = (0.52E_0 - 0.09) \text{ g/cm}^2$$

Hence, the endpoint energy for Sr-90 is given by:

$$E_0 = \frac{R + 0.09}{0.52}$$

Using the calculated range $R_{Sr} = 0.8444g/cm^2$, the endpoint energy was obtained as:

$$E_0^{(\mathrm{Sr})} = 1.80 \text{ MeV}$$

The half thickness method provides a measure of the beta particle range. The endpoint energy calculated for Sr-90 is close to theoretical value of endpoint energy $\approx$ 2.28MeV, demonstrating the dependence of particle range on energy and validating the attenuation behavior of beta particles in matter.

F. Back-scattering of β particles

The back-scattering of beta particles from an aluminum absorber was studied by measuring the count rate as a function of absorber thickness. The net count rate was obtained after background subtraction and normalization over the counting time.

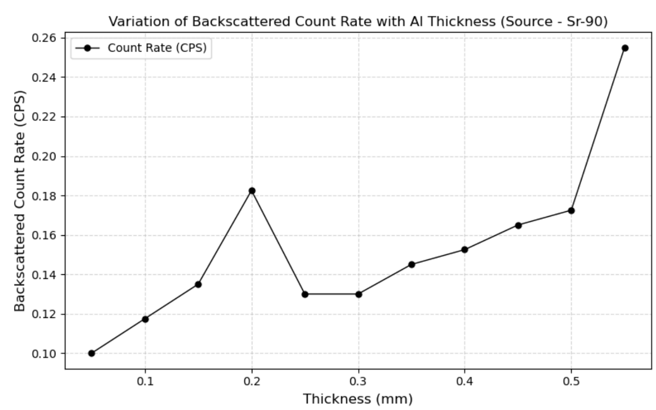

FIG. 14. Variation of back-scattered count rate with absorber thickness (Sr-90 source).

It is observed that the backscattered count rate increases with absorber thickness (see Figure 14). This is due to the increased probability of beta particles undergoing scattering interactions within the material and being redirected towards the detector.

However, the overall count rate is relatively low, leading to noticeable statistical fluctuations in the data. Despite this, the general trend of increasing count rate with thickness is clearly observed and is consistent with the expected behavior of beta particle back-scattering.

G. Production & Attenuation of Bremsstrahlung

TABLE I. Observation Table for Production and Attenuation of Bremsstrahlung

S.No	Absorber Position	Counts	Net Counts
For Al (1.8mm) & Perspex (1.8mm) combination			
1	No Absorber	9149	8805
2	Perspex facing source	814	470
3	Al facing source	821	477
For Perspex (1.8mm) & Cu (0.3mm) combination			
1	Cu facing source	710	366
2	Perspex facing source	645	304
For Al (1.8mm) & Cu (0.3mm) combination			
1	Al facing source	634	290
2	Cu facing source	636	292

The data demonstrates that Bremsstrahlung production depends on the atomic number (Z) of the target. For every tested combination, placing the higher-Z material facing the source resulted in a higher net count rate. This confirms the theoretical principle that Bremsstrahlung intensity increases strongly with the atomic number of the absorbing medium.

A high-Z material facing the source produces an intense Bremsstrahlung flux that easily passes through a subsequent low-Z shield. Conversely, a low-Z material facing the source generates weak Bremsstrahlung, which is then absorbed by the high-Z backing. This proves that for proper shielding of beta emitters, a low-Z material must be placed first to stop beta particles with minimal X-ray production, followed by a high-Z material to absorb any resulting secondary radiation.

V. ERROR ANALYSIS

A. Uncertainty in the slope of plateau region of GM characteristics

The percentage slope of the plateau is given by:

$$\%\text{slope} = \frac{m \times 100}{N_1} \times 100$$

where m is the slope of the plateau and N_1 is the count rate at the beginning of the plateau.

Let

$$S = \frac{100^2\, m}{N_1}$$

The uncertainty in S is obtained using error propagation:

$$\Delta S = \sqrt{\left(\frac{\partial S}{\partial m}\Delta m\right)^2 + \left(\frac{\partial S}{\partial N_1}\Delta N_1\right)^2}$$

We have,

$$\Delta S = \sqrt{\left(\frac{100^2}{N_1}\Delta m\right)^2 + \left(\frac{100^2 m}{N_1^2}\Delta N_1\right)^2}$$

Hence, the percentage uncertainty is:

$$\frac{\Delta S}{S} = \sqrt{\left(\frac{\Delta m}{m}\right)^2 + \left(\frac{\Delta N_1}{N_1}\right)^2}$$

Using the fitted values and $\Delta N_1 = \sqrt{N_1}$, the percentage slope and its uncertainty were obtained as:

$$\Delta(\%\text{slope})_\gamma = 1.4\%$$
$$\Delta(\%\text{slope})_\beta = 2.3\%$$

B. Uncertainty in Efficiency

The efficiency is given by:

$$\eta(\%) = \frac{N}{R} \times 100 \tag{15}$$

Using error propagation:

$$\Delta\eta = \sqrt{\left(\frac{\partial \eta}{\partial N}\Delta N\right)^2 + \left(\frac{\partial \eta}{\partial R}\Delta R\right)^2}$$

Thus, the fractional uncertainty is obtained as:

$$\frac{\Delta\eta}{\eta} = \sqrt{\left(\frac{\Delta N}{N}\right)^2 + \left(\frac{\Delta R}{R}\right)^2}$$

Here, $\Delta N = \sqrt{\Sigma_i n_i}/(6t)$ (using Poisson statistics), where t = time of acquisition and n_i are the counts summed over in presence of source and background. ΔR is obtained from propagation of uncertainty in distance from source and the diameter of detector window, which is given by the expression:

$$\Delta R = \frac{Ad^2}{8D^2}\sqrt{\left(\frac{\Delta d}{d}\right)^2 + \left(\frac{\Delta D}{D}\right)^2}$$

$$\frac{\Delta R}{R} = 2\sqrt{\left(\frac{\Delta d}{d}\right)^2 + \left(\frac{\Delta D}{D}\right)^2}$$

Using $\Delta d = \Delta D = 0.1cm$ and Uncertainty in R,

$$\text{For } \gamma \text{ source}, \Delta R_\gamma = 24.5$$
$$\text{For } \beta \text{ source}, \Delta R_\beta = 27.0$$

Using the above relations, the efficiencies with uncertainties were obtained as:

$$\Delta(\eta_\beta) = 0.69\% \tag{16}$$
$$\Delta(\eta_\gamma) = 0.09\% \tag{17}$$

The uncertainty in efficiency is caused by statistical fluctuations in count rate and uncertainties in source activity and distance measurements.

C. Uncertainty in Endpoint energy and Range of β particle

The range of Sr-90 is given by the equation (4). Using error propagation, the fractional uncertainty in R_{Sr} is:

$$\frac{\Delta R_{\mathrm{Sr}}}{R_{\mathrm{Sr}}} = \sqrt{\left(\frac{\Delta t_{1/2}^{(\mathrm{Sr})}}{t_{1/2}^{(\mathrm{Sr})}}\right)^2 + \left(\frac{\Delta t_{1/2}^{(\mathrm{Tl})}}{t_{1/2}^{(\mathrm{Tl})}}\right)^2}$$

The half-thickness ($t_{1/2}$) is determined via linear interpolation between the data points (x_1, y_1) and (x_2, y_2) bracketing the half-maximum count ($y_{half} = y_{max}/2$). Assuming exact absorber thicknesses, the uncertainty in $t_{1/2}$ stems solely from the Poisson error of the maximum count. Mapping this vertical statistical error ($\Delta y_{half} = \sqrt{y_{max}}/2$) to the horizontal axis using the interpolation slope (m) yields:

$$\Delta t_{1/2} = \left|\frac{\Delta y_{half}}{m}\right| = \left|\frac{x_2 - x_1}{y_2 - y_1}\right| \frac{\sqrt{y_{max}}}{2}$$

The endpoint energy is given by the equation (3). Hence, the uncertainty in E_0 is:

$$\Delta E_0 = \frac{\Delta R}{0.52}$$

Using uncertainty in $t_{1/2}, R$ and E_0 are calculated from above formulae to be:

$$\text{For Tl-204}, t_{1/2}^{(Tl)} = 7.7mg/cm^2$$
$$\text{For Sr-90}, t_{1/2}^{(Sr)} = 25.5mg/cm^2$$
$$\text{Range of } \beta \text{ particle}, \Delta R = 0.27g/cm^2$$
$$\text{End-point energy}, \Delta(E_0) = 0.52MeV$$

D. Least squares Linear fit

We fit the observed data-points into a form,

$$y = mx + c$$

The statistical error calculated in linear fit is done using the following formulae [3],

$$\sigma_y = \sqrt{(\sum_i \frac{(\bar{y} - y_i)^2}{N-2})} \tag{18}$$

Errors in slope and intercept,

$$\Delta m = \sigma_y \sqrt{\frac{\sum_i x^2}{((N\sum_i x^2) - (\sum_i x)^2)}} \tag{19}$$

$$\Delta c = \sigma_y \sqrt{\frac{N}{((N\sum_i x^2) - (\sum_i x)^2)}} \tag{20}$$

All code related to the plots, table and calculations is available on this Github Repository [4].

VI. RESULTS AND DISCUSSION

A. GM Characteristics

The GM counter characteristic curves for both γ and β sources exhibit a well-defined plateau in the range 360 V – 600 V. The slopes of the plateau were obtained as $m_\gamma = 1.65 \pm 0.29$ and $m_\beta = 0.36 \pm 0.07$, corresponding to percentage slopes of $(7.8 \pm 1.4)\%$ and $(11.6 \pm 2.3)\%$, respectively. The operating voltage was chosen in the middle of the plateau as $V_{\mathrm{op}} = 480$ V.

B. Inverse square law

The inverse square law was verified by fitting the count rate as $R = A + \frac{B}{d^2}$, yielding $A = 20.84 \pm 5.38$ and $B = 304.79 \pm 30.52$. The parameter A represents background contribution, while B reflects source strength. The log-log fit gave a slope $n = -1.49 \pm 0.07$, which deviates slightly from the ideal value of -2 due to finite detector size, scattering effects, and alignment errors. Nevertheless, the agreement within uncertainty confirms the expected inverse square dependence of radiation intensity.

C. Efficiency of GM detector

The present activities of the sources were calculated as $A_\beta = 1.60$ kBq and $A_\gamma = 62.7$ kBq. The incident disintegration rates were $R_\gamma = 352.7$ and $R_\beta = 225.0$. The efficiencies were obtained as $\eta_\beta = (5.47 \pm 0.69)\%$ and $\eta_\gamma = (1.34 \pm 0.09)\%$. The significantly higher efficiency for β particles arises from their greater ionization capability compared to γ rays, which interact weakly and often pass through the detector without interaction. These values are consistent with the expected behavior of GM detectors.

D. Nuclear counting statistics

For background radiation, the mean count was $\bar{N} = 70.56$, variance $\sigma^2 = 88.52$, and standard deviation $\sigma = 9.41$, showing reasonable agreement with Poisson statistics where $\sigma^2 \approx \bar{N}$. The Gaussian fit yielded $\mu = (72.75 \pm 1.27)$ and $\sigma = (9.61 \pm 1.29)$, further confirming this behavior. For β source counts, $\bar{N} = 734.55$ and $\sigma^2 = 700.86$, again almost $\sigma^2 \approx \bar{N}$. The normalized distribution gave $\mu = (0.32 \pm 0.04)$ and $\sigma = (1.00 \pm 0.04)$, indicating proper normalization and convergence to Gaussian distribution at high counts, as predicted.

E. Endpoint energy of β particle

In the half-thickness method, the values obtained were $t_{1/2}^{(\mathrm{Tl})} = 37.3$ mg/cm^2 and $t_{1/2}^{(\mathrm{Sr})} = 102.5$ mg/cm^2. The range of Sr-90 was calculated as $R_{\mathrm{Sr}} = 0.8444$ g/cm^2 with uncertainty $\Delta R = 0.27$ g/cm^2. The endpoint energy was determined as $E_0^{(\mathrm{Sr})} = 1.80 \pm 0.52$ MeV, which is reasonably close to the theoretical value of ~ 2.28 MeV. The discrepancy may arise from energy loss in air, absorber non-uniformity, and interpolation errors in determining half-thickness.

F. Back-scattering of β particles

The back-scattering of β particles was observed to increase with absorber thickness, indicating a higher probability of scattering interactions within the material. However, the overall count rates were relatively low, leading to significant statistical fluctuations in the data. These fluctuations arise due to limited counting statistics and the random nature of scattering events. Despite this, the general increasing trend is clearly visible and agrees with the expected behavior of back-scattering with increasing thickness.

G. Production and Attenuation of Bremsstrahlung

The Bremsstrahlung analysis shows that higher-Z materials produce larger count rates when placed facing the β source, confirming that radiation intensity increases with atomic number. This is attributed to stronger deceleration of charged particles in high-Z nuclei, leading to enhanced X-ray production. It is also observed that placing a low-Z material before a high-Z absorber reduces the detected radiation. This is consistent with theoretical expectations.

VII. CONCLUSION

The GM counter was successfully characterized, with an optimal operating voltage of 480 V and stable plateau behavior. The inverse square law was verified with an experimental exponent $n = -1.49 \pm 0.07$. The detector efficiencies were determined as $(5.47 \pm 0.69)\%$ for β and $(1.34 \pm 0.09)\%$ for γ radiation, highlighting the dependence on interaction mechanisms.

Statistical analysis confirmed that radioactive decay follows Poisson statistics, transitioning to Gaussian behavior at high count rates. The endpoint energy of Sr-90 was estimated as 1.80 ± 0.52 MeV using the half-thickness method, in agreement with theory.

Additionally, the experiments on back-scattering and Bremsstrahlung validated their dependence on absorber thickness and atomic number, respectively. Thus, the experiment successfully demonstrates key principles of radiation detection, statistical nature of decay, and interaction of radiation with matter. Slight deviations in some experiments can be attributed to the sources of error mentioned below.

VIII. SOURCES OF ERROR

1. **Statistical fluctuations:** Radioactive decay is inherently random, leading to Poisson fluctuations in the measured count rates, especially significant in low-count measurements such as back-scattering.

2. **Detector dead time:** At higher count rates, the finite dead time of the GM counter leads to undercounting, causing deviation from true count rates.

3. **Alignment errors:** Improper alignment of source, absorber, and detector can affect attenuation, back-scattering, and Bremsstrahlung measurements.

[1] National Institute of Science Education and Research, Bhubaneswar. *Basics of GM Counter*. Lab Manual.

[2] National Institute of Science Education and Research, Bhubaneswar. *Applications of GM Counter*. Lab Manual.

[3] John R. Taylor. *An Introduction to Error Analysis: The Study of Uncertainties in Physical Measurements*. University Science Books, Sausalito, CA, 2nd edition, 1997.

[4] Aryan Shrivastava. P341 - nuclear physics and instrumentation lab. `https://github.com/crimsonpane23/P341-Nuclear-Physics-and-Instrumentation-Lab.git`, 2026.